

Problem

For $F(\omega) = \frac{3}{3 + j\omega}$, find $J = \int_{-\infty}^{\infty} f^2(t) dt$.

Step-by-step solution

Step 1 of 2

The function is, $F(\omega) = \frac{1}{3 + j\omega}$.

Calculate the value of $J = \int_{-\infty}^{\infty} f^2(t) dt$.

$$J = \int_{-\infty}^{\infty} f^2(t) dt$$

Changing the limits,

$$J = 2 \int_0^{\infty} f^2(t) dt$$

Change the time domain to frequency domain,

$$\begin{aligned} J &= \frac{2}{2\pi} \int_0^{\infty} |F(\omega)|^2 d\omega \\ &= \frac{1}{\pi} \int_0^{\infty} \left| \frac{1}{(3 + j\omega)} \right| d\omega \\ &= \frac{1}{\pi} \int_0^{\infty} \left(\frac{1}{3 + j\omega} \right) \left(\frac{1}{3 - j\omega} \right) d\omega \\ &= \frac{1}{\pi} \int_0^{\infty} \frac{1}{(9 + \omega^2)} d\omega \end{aligned}$$

Step 2 of 2

It is known that,

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$$

Therefore,

$$\begin{aligned} J &= \frac{1}{3\pi} \tan^{-1} \left(\frac{\omega}{3} \right) \\ &= \frac{1}{3\pi} \times \frac{\pi}{2} \\ &= \frac{1}{6} \end{aligned}$$

Therefore, the value of J is $\boxed{\frac{1}{6}}$.